

Experiments 0–1: Width Validation & Centralized Phase Boundary

Federated Learning Grokking Project

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1 Experiment 0: Width Validation

Goal. Verify that hidden width $N = 256$ supports grokking across the α range, so that grokking failures in later FL experiments are attributable to federation—not insufficient model capacity.

Setup.

- Task: $(a + b) \bmod 97$, quadratic activation, MSE loss, full-batch GD, $\text{lr} = 50$
- $N \in \{100, 128, 200, 256\}$, $\alpha \in \{0.1, 0.3, 0.5\}$, seed = 42
- $S_{\max} = 50,000$ gradient steps per run (12 runs total)

Key findings.

- **Width does not matter** in this regime. All four widths produce nearly identical T_{grok} at matched α (Table 1).
- At $\alpha = 0.1$, **no width groks** within 50,000 steps. Extended runs to 200,000 steps confirm: test accuracy stays at $\sim 0.5\%$, IPR stays at ~ 0.02 (random-init level), and test loss is *monotonically rising*. This is not a budget issue—the model is actively diverging from generalisation.
- At $\alpha \geq 0.3$, all widths reach 100% test accuracy with IPR ~ 0.42 – 0.49 .
- **Conclusion:** $N = 256$ is validated as the fixed width for all subsequent experiments. The interesting variable is α , not N .

Table 1: Experiment 0: T_{grok} across widths and training fractions.

α	$N = 100$	$N = 128$	$N = 200$	$N = 256$
0.1	∞	∞	∞	∞
0.3	13,100	12,200	12,200	12,600
0.5	8,600	8,300	7,800	7,600

Experiment 0: Width Validation ($p=97$, GD, $lr=50$, 50k steps)

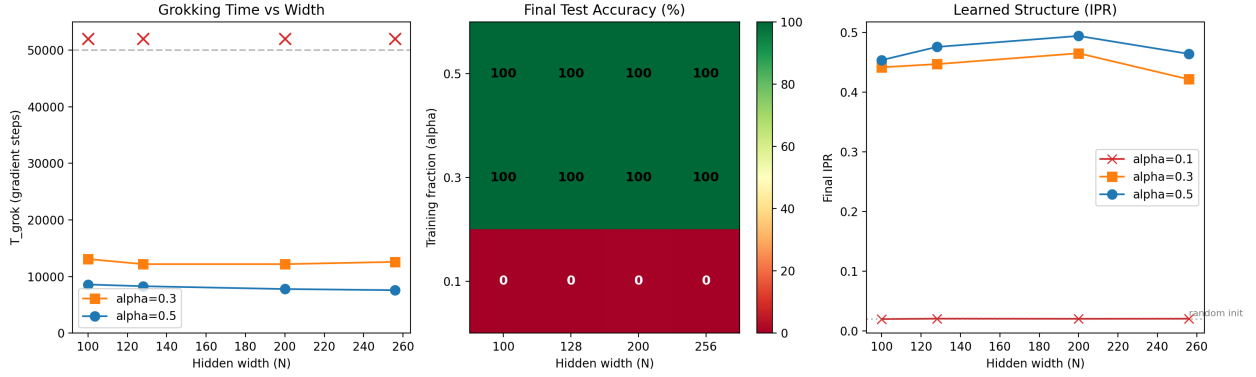


Figure 1: Experiment 0 summary. **Left:** T_{grok} vs. width—flat at each α ; red crosses indicate no grokking. **Centre:** Final test accuracy heatmap; clean binary transition between $\alpha = 0.1$ (0%) and $\alpha = 0.3$ (100%). **Right:** Final IPR confirms Fourier structure only forms when grokking occurs.

Experiment 0: Test Accuracy Curves ($p=97$, GD, $lr=50$)

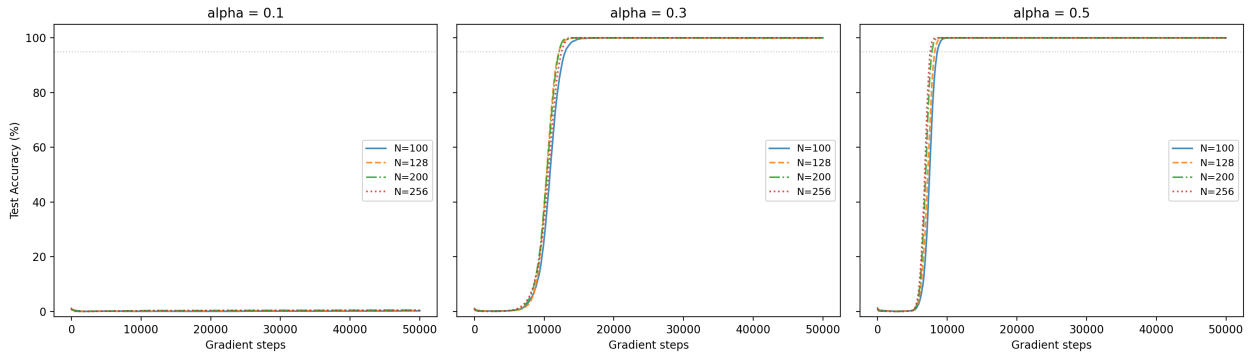


Figure 2: Test accuracy curves for all Exp 0 runs. At $\alpha = 0.3$ and 0.5 , the classic grokking S-curve is visible, with all widths overlapping. At $\alpha = 0.1$, test accuracy is indistinguishable from chance ($1/97 \approx 1\%$).

2 Experiment 1: Centralized Phase Boundary

Goal. Determine the critical training fraction α_c below which centralised grokking fails, and measure $T_{\max}$ (the slowest grokking observed) to calibrate all FL step budgets.

Setup.

- $N = 256$ (validated in Exp 0), all other hyperparameters as in Exp 0
- $\alpha \in \{0.03, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3, 0.5\}$
- Seeds $\in \{42, 123, 456\}$ (3 seeds per α)
- $S_{\max} = 100,000$ steps (24 runs)

Phase boundary results (at $S_{\max} = 100,000$).

- $\alpha \leq 0.2$: **No grokking**. All seeds fail. Test accuracy $< 7\%$ at best.
- $\alpha = 0.3$: **3/3 seeds grok**. $T_{\text{grok}} = 12,833 \pm 404$ steps.
- $\alpha = 0.5$: **3/3 seeds grok**. $T_{\text{grok}} = 7,667 \pm 115$ steps.
- Initial estimate: $\alpha_c^{\text{practical}} = 0.3$, $T_{\max} = 13,300$.

Table 2: Experiment 1: Phase boundary at $S_{\max} = 100,000$.

α	n_{train}	Grokked	T_{grok} (mean $\pm$ std)	Final acc (%)	Final IPR
0.03	282	0/3	∞	0.0	0.020
0.05	470	0/3	∞	0.1	0.020
0.075	705	0/3	∞	0.3	0.021
0.1	940	0/3	∞	0.5	0.021
0.15	1,411	0/3	∞	0.6	0.022
0.2	1,881	0/3	∞	3.5	0.054
0.3	2,822	3/3	$12,833 \pm 404$	100.0	0.472
0.5	4,704	3/3	$7,667 \pm 115$	100.0	0.484

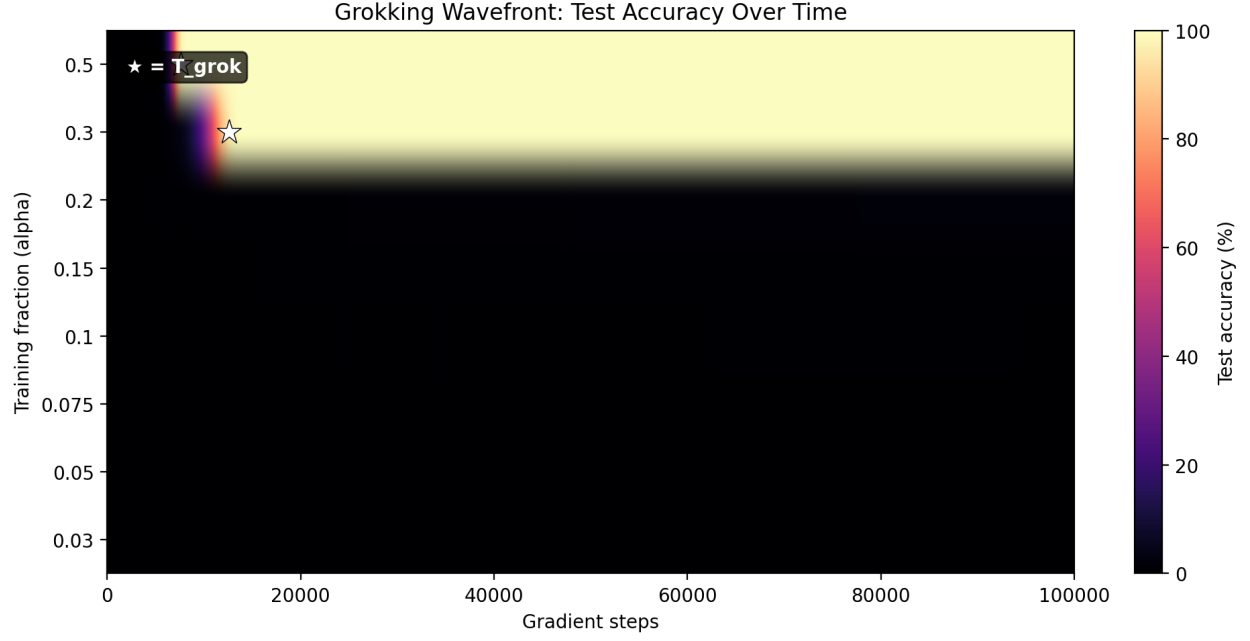


Figure 3: Grokking wavefront. Each row is one α value; colour encodes test accuracy over time. The sharp horizontal edge between $\alpha = 0.2$ and 0.3 is the phase boundary. Stars mark T_{grok} .

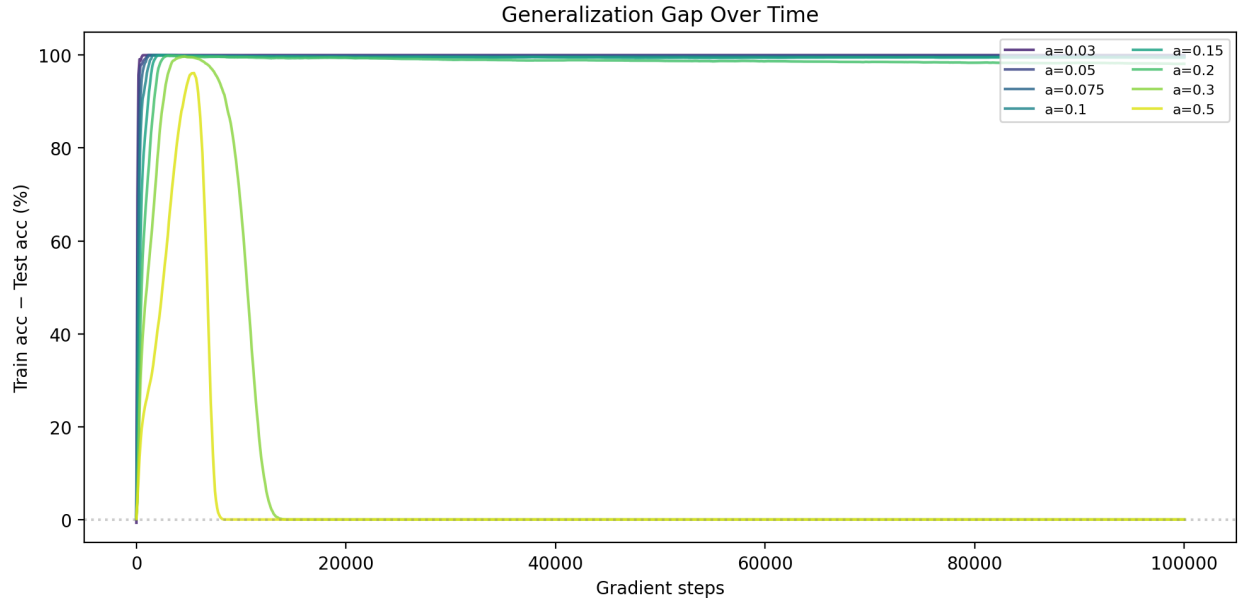


Figure 4: Generalisation gap (train acc – test acc) over time. All α values memorise by $\sim 2,000$ steps (gap $\rightarrow 100\%$). For $\alpha \geq 0.3$, the gap collapses as grokking completes. For $\alpha \leq 0.2$, the gap remains at 100% permanently.

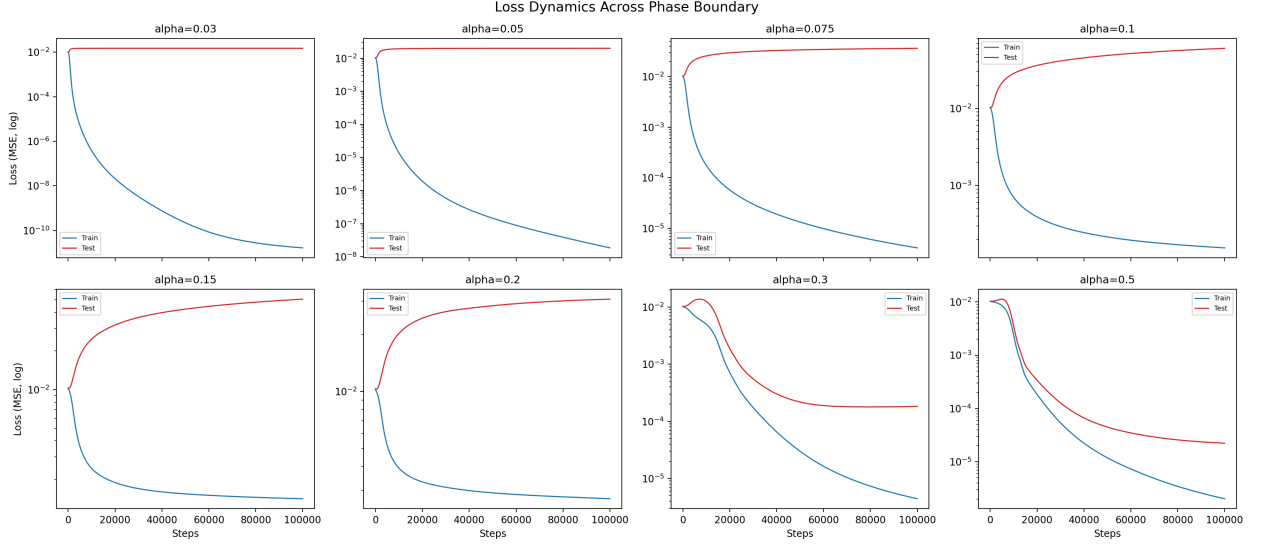


Figure 5: Train and test loss (log scale) for each α . Below the boundary ($\alpha \leq 0.15$), test loss rises monotonically. At $\alpha = 0.2$, test loss plateaus. At $\alpha \geq 0.3$, test loss eventually plunges—the grokking moment.

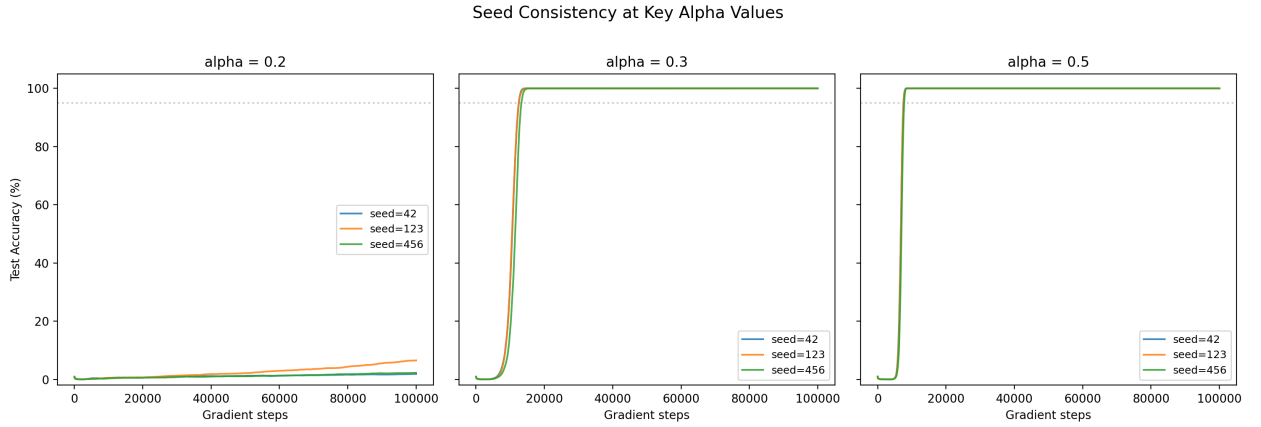


Figure 6: Seed consistency at key α values. At $\alpha = 0.2$, all 3 seeds agree: no grokking. At $\alpha = 0.3$, all 3 grok within 700 steps of each other. The phenomenon is highly reproducible.

2.1 Extended Runs: Refining the True Phase Boundary

Motivation. The $\alpha = 0.2$ runs at 100,000 steps showed test accuracy of 3.5% and IPR of 0.054—above random init (0.02) but far from grokked (~ 0.47). Is this a true boundary, or is grokking merely very slow?

Result. Extended to 1,000,000 steps, $\alpha = 0.2$ **groks**:

- $T_{50} = 555,000$, $T_{\text{grok}} = 805,000$
- Final test accuracy: 98.7%, final IPR: 0.253
- The S-curve shape is *identical* to $\alpha = 0.3$ and 0.5 , stretched by two orders of magnitude

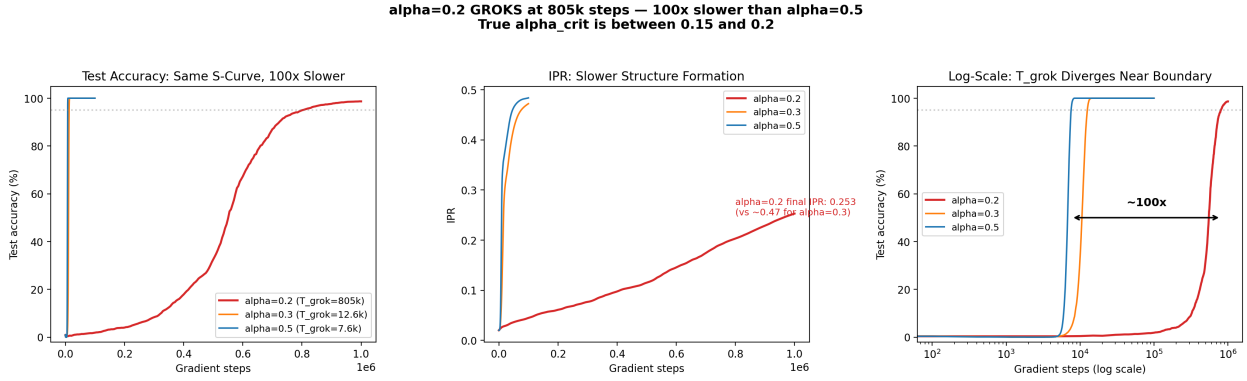


Figure 7: $\alpha = 0.2$ groks at $T_{\text{grok}} = 805,000$ —roughly $100\times$ slower than $\alpha = 0.5$. **Left:** Test accuracy curves for $\alpha \in \{0.2, 0.3, 0.5\}$. **Centre:** IPR at $\alpha = 0.2$ is still climbing at 1M steps. **Right:** Log-scale view reveals the $\sim 100\times$ gap.

Fine-grained boundary scouting. Additional runs at $\alpha \in \{0.21, 0.22, 0.25, 0.28\}$ reveal the full scaling:

Table 3: Refined phase boundary: T_{grok} vs. α and dataset size.

α	n_{train}	T_{50}	T_{grok}	Ratio vs. $\alpha=0.5$	Budget needed
0.20	1,881	555,000	805,000	$106\times$	$\sim 1\text{M}$
0.21	1,975	138,000	166,000	$22\times$	$\sim 200\text{k}$
0.22	2,069	45,000	59,000	$7.8\times$	$\sim 80\text{k}$
0.25	2,352	19,000	24,000	$3.2\times$	$\sim 40\text{k}$
0.28	2,634	13,000	16,000	$2.1\times$	$\sim 25\text{k}$
0.30	2,822	10,600	12,600	$1.7\times$	$\sim 20\text{k}$
0.50	4,704	6,800	7,600	$1.0\times$	$\sim 12\text{k}$

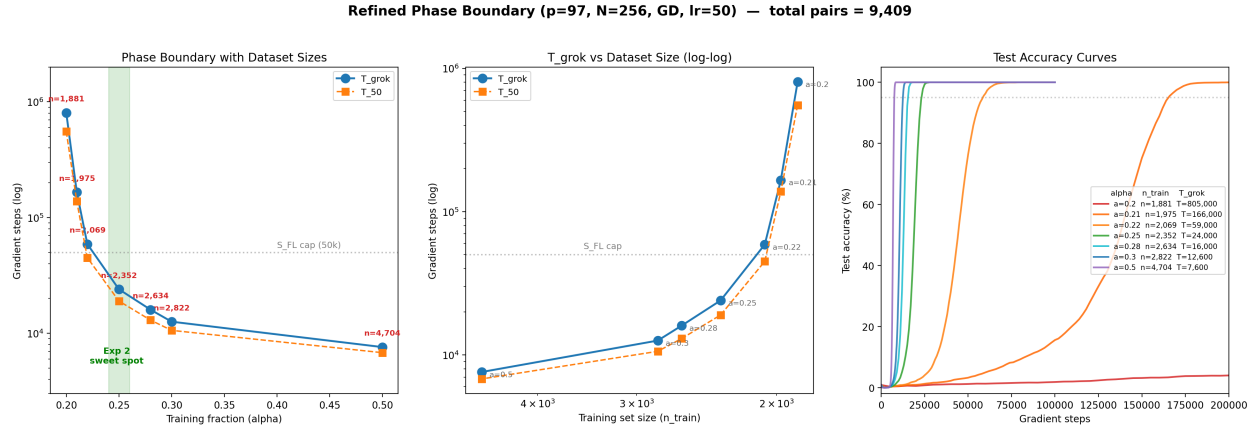


Figure 8: Refined phase boundary with dataset sizes annotated. **Left:** T_{grok} vs. α on log scale; the green band marks $\alpha = 0.25$, the practical sweet spot for FL experiments. **Centre:** T_{grok} vs. n_{train} on log-log axes. **Right:** Test accuracy curves zoomed to 200,000 steps.

2.2 Mechanistic Analysis: Critical Scaling

Key finding. The grokking step follows a **power-law divergence**:

$$T_{\text{grok}} \propto (\alpha - \alpha_c)^{-\gamma}, \quad \alpha_c \approx 0.198, \quad \gamma \approx 0.99, \quad R^2 = 0.978 \quad (1)$$

Interpretation.

- The exponent $\gamma \approx 1$ means T_{grok} is roughly *inversely proportional* to the distance from the critical point. This explains the extreme sensitivity: adding 94 training samples ($\alpha : 0.20 \rightarrow 0.21$) cuts T_{grok} by $5\times$ because the denominator $(\alpha - \alpha_c)$ grows from 0.002 to 0.012.
- $\alpha_c \approx 0.198$ corresponds to $n_{\text{train}} \approx 1,863$ samples out of $p^2 = 9,409$ total pairs, or a parameter-to-data ratio of ~ 40 .
- This is a **second-order phase transition**: no discontinuity in the order parameter (test accuracy at fixed time), only a diverging timescale.

Mechanistic picture (Figure 9).

- **All α values traverse the same dynamical path** through (IPR, test accuracy) space. Grokking is a universal process; α only controls the clock speed.
- **Memorise first, then organise.** Train loss drops immediately (memorisation), but IPR stays flat. Only after memorisation is complete does the network begin reorganising weights into Fourier structure.
- **Less data = weaker gradient signal for the Fourier solution.** The memorised solution provides less gradient information pointing toward the correct periodic structure. Near α_c , this signal is so weak that the optimisation takes exponentially longer to separate signal from noise in weight space.
- **Test loss inflection marks grokking onset.** The peak of test loss (before it starts declining) is the precise moment the Fourier solution begins to dominate. This inflection shifts from $\sim 5,000$ steps ($\alpha = 0.5$) to $> 200,000$ steps ($\alpha = 0.20$).

Mechanistic Investigation: Why Does T_{grok} Diverge at the Phase Boundary?

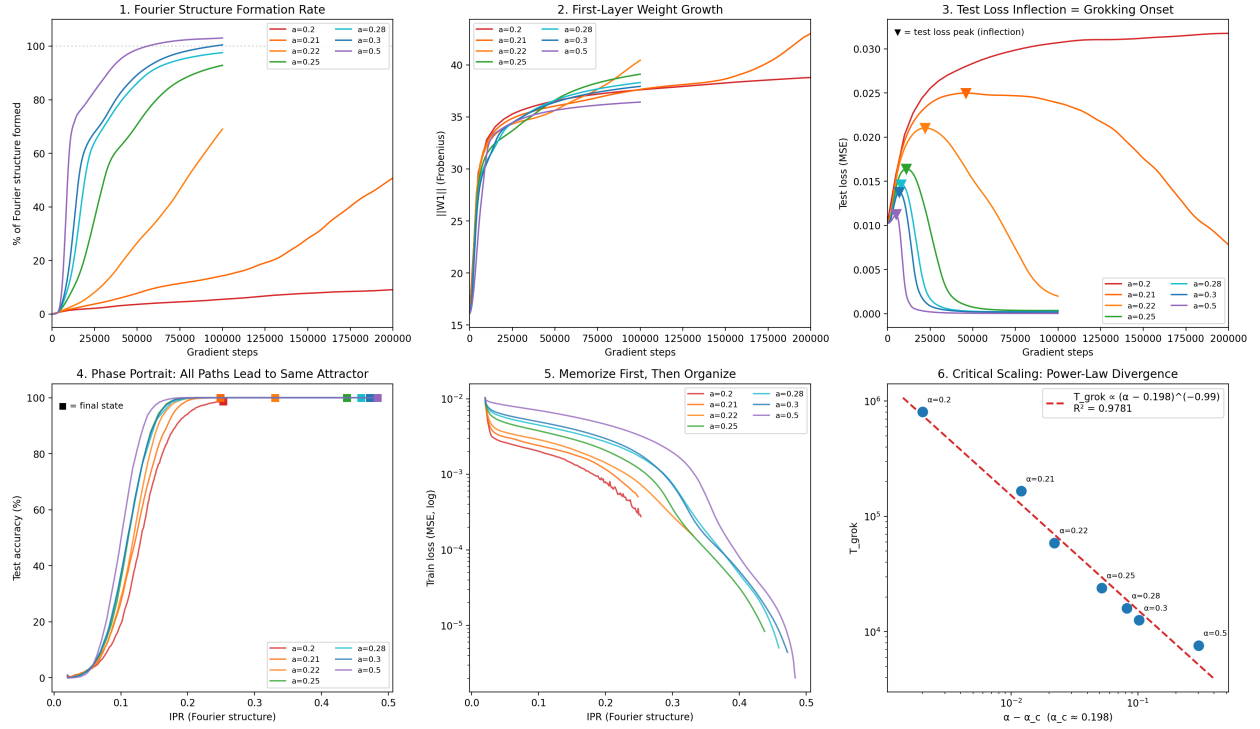


Figure 9: Mechanistic investigation of the phase boundary. **(1)** Fourier structure formation rate: all α build the same representation at different speeds. **(2)** First-layer weight growth drives grokking. **(3)** Test loss inflection point = grokking onset. **(4)** Phase portrait: all trajectories follow the same attractor. **(5)** Train loss vs. IPR: memorisation precedes organisation. **(6)** Power-law fit: $T_{\text{grok}} \propto (\alpha - 0.198)^{-0.99}$, $R^2 = 0.978$.

3 Implications for FL Experiments

- **True critical point:** $\alpha_c \approx 0.198$ (not 0.3 as the 100,000-step budget suggested). The practical boundary depends on compute budget.
- **FL sweet spot:** $\alpha = 0.25$ ($T_{\text{grok}} = 24,000$, $n_{\text{train}} = 2,352$). An FL budget of $S_{\text{FL}} = 50,000$ gives $2\times$ headroom—tight enough that FL overhead matters, generous enough that grokking is achievable.
- **Experiment 2 grid:** $\alpha \in \{0.25, 0.3, 0.5\}$, $K \in \{2, 5, 10, 20\}$. At $\alpha = 0.25$ with $K = 5$: each client sees ~ 470 samples ($\alpha_{\text{eff}} = 0.05$ per client), which is well below the centralised phase boundary. If FedAvg groks here, aggregation is powerful.
- **Adaptive budgets:** $S_{\text{FL}} = 50,000$ (cap), $S_{\text{rescue}} = 80,000$ (cap).
- **The central question for Exp 2:** Does FL shift α_c ? If centralised groks at $\alpha = 0.25$ in 24,000 steps but FL does not grok within 50,000, that is a measurable shift of the phase boundary.